Laboratory Record

of COMPUTER NETWORKS LAB

Roll No.: 1601-24-733-

Experiment No.: Lab Task - 7

Sheet No.:

Date.:

LAB TASK – 7

Aim: Simulation of Stop and Wait protocol

Description :

Stop-and-Wait is a simple **flow-control and error-control protocol**.
The sender transmits **one frame at a time** and then waits for an acknowledgment (ACK).
The sender sends the next frame only after receiving the ACK for the previous frame.
If the ACK is not received within a specified time, the sender assumes that the frame or ACK was lost and **retransmits the frame**. However, waiting for an ACK after every frame can reduce network efficiency, especially when the propagation delay is high.

Procedure : Open Ubuntu-24.04 Terminal.

- cd ns-allinone-3.42/
- cd ns-3.42/
- sudo nano scratch/stopnwait.cc
- ./ns3 run stopnwait
  (or)
- ./ns3 run scratch/stopnwait
- exit

```
student@cselab3-06: ~/ns-all
student@cselab3-06:~$ cd ns-allinone-3.42/
student@cselab3-06:~/ns-allinone-3.42$ cd ns-3.42/
student@cselab3-06:~/ns-allinone-3.42/ns-3.42$ sudo nano scratch/stopnwait.cc
[sudo] password for student:
student@cselab3-06:~/ns-allinone-3.42/ns-3.42$ ./ns3 run stopnwait
-- Using default output directory /home/student/ns-allinone-3.42/ns-3.42/build
-- CCache is enabled.
-- Proceeding without cmake-format
-- find_external_library: SQLite3 was found.
```

- cd ns-allinone-3.42/
- cd netanim-3.109/
- ./NetAnim

```
student@cselab3-06: ~/ns-all
student@cselab3-06:~$ cd ns-allinone-3.42/
student@cselab3-06:~/ns-allinone-3.42$ cd netanim-3.109/
student@cselab3-06:~/ns-allinone-3.42/netanim-3.109$ ./NetAnim
```

1) Now the NetAnim window opens.
2) At the top-left corner click on the folder symbol (explorer)
3) Click on Student -> ns-allinone-3.42 -> ns-3.42 -> stop-and-wait.xml
4) Select that xml file and click on OPEN.
5) Now to start the simulation: Click on PLAY button on top-left.
6) The simulation starts…

Code : stopnwait.cc

```
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"
#include "ns3/netanim-module.h"

using namespace ns3;

Ptr<Socket> s, r;
bool ack = true;
int seq = 0, sent = 0, maxP = 5;

void Send()
{
    if (sent >= maxP) return;

    if (ack) {
        uint8_t d = 0;
        s->Send(Create<Packet>(&d, 1));
        NS_LOG_UNCOND("Sender: DATA " << seq);
        ack = false;
    } else {
        uint8_t d = 0;
        s->Send(Create<Packet>(&d, 1));
        NS_LOG_UNCOND("Sender: Retransmit DATA " << seq);
    }

    Simulator::Schedule(Seconds(2), &Send);
}

void Receive(Ptr<Socket> socket)
{
    Ptr<Packet> p;
    while ((p = socket->Recv())) {
        uint8_t type;
        p->CopyData(&type, 1);

        if (type == 0) {
            NS_LOG_UNCOND("Receiver: DATA " << seq);
            uint8_t a = 1;
            r->Send(Create<Packet>(&a, 1));
            NS_LOG_UNCOND("Receiver: ACK " << seq);
            seq++;
            sent++;
        }
```

```
    else {
        NS_LOG_UNCOND("Sender: ACK received");
        ack = true;
      }
   }
}

int main()
{
   NodeContainer n;
   n.Create(2);

   PointToPointHelper p;

   p.SetDeviceAttribute("DataRate", StringValue("5Mbps"));
   p.SetChannelAttribute("Delay", StringValue("2ms"));

   NetDeviceContainer d = p.Install(n);

   InternetStackHelper stack;
   stack.Install(n);

   Ipv4AddressHelper ip;

   ip.SetBase("10.1.1.0", "255.255.255.0");
   Ipv4InterfaceContainer i = ip.Assign(d);

   s = Socket::CreateSocket(n.Get(0), UdpSocketFactory::GetTypeId());
   r = Socket::CreateSocket(n.Get(1), UdpSocketFactory::GetTypeId());

   s->Bind(InetSocketAddress(Ipv4Address::GetAny(), 8080));
   r->Bind(InetSocketAddress(Ipv4Address::GetAny(), 8080));

   s->Connect(InetSocketAddress(i.GetAddress(1), 8080));
   r->Connect(InetSocketAddress(i.GetAddress(0), 8080));

   s->SetRecvCallback(MakeCallback(&Receive));
   r->SetRecvCallback(MakeCallback(&Receive));

   AnimationInterface anim("stop-and-wait.xml");

   anim.SetConstantPosition(n.Get(0), 10, 30);
   anim.SetConstantPosition(n.Get(1), 50, 30);

   Simulator::Schedule(Seconds(1), &Send);
   Simulator::Stop(Seconds(15));
   Simulator::Run();
   Simulator::Destroy();
}
```

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